

```
In [26]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [3]: # Load the data set
df=pd.read_csv(r"C:\Users\abhis\Downloads\WA_Fn-UseC_-HR-Employee-Attrition.csv")
```

First look at the data

```
In [5]: df.head(5)
```

Out[5]:

	Age	Attrition	BusinessTravel	DailyRate	Department	Distance
0	41	Yes	Travel_Rarely	1102	Sales	1
1	49	No	Travel_Frequently	279	Research & Development	8
2	37	Yes	Travel_Rarely	1373	Research & Development	2
3	33	No	Travel_Frequently	1392	Research & Development	3
4	27	No	Travel_Rarely	591	Research & Development	2

5 rows × 35 columns



```
In [6]: df.shape
```

Out[6]: (1470, 35)

```
In [7]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1470 entries, 0 to 1469
Data columns (total 35 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Age                                   1470 non-null   int64
1   Attrition                           1470 non-null   object
2   BusinessTravel                      1470 non-null   object
3   DailyRate                           1470 non-null   int64
4   Department                          1470 non-null   object
5   DistanceFromHome                   1470 non-null   int64
6   Education                           1470 non-null   int64
7   EducationField                      1470 non-null   object
8   EmployeeCount                      1470 non-null   int64
9   EmployeeNumber                     1470 non-null   int64
10  EnvironmentSatisfaction              1470 non-null   int64
11  Gender                              1470 non-null   object
12  HourlyRate                          1470 non-null   int64
13  JobInvolvement                      1470 non-null   int64
14  JobLevel                            1470 non-null   int64
15  JobRole                             1470 non-null   object
16  JobSatisfaction                     1470 non-null   int64
17  MaritalStatus                      1470 non-null   object
18  MonthlyIncome                      1470 non-null   int64
19  MonthlyRate                        1470 non-null   int64
20  NumCompaniesWorked                 1470 non-null   int64
21  Over18                             1470 non-null   object
22  OverTime                           1470 non-null   object
23  PercentSalaryHike                  1470 non-null   int64
24  PerformanceRating                  1470 non-null   int64
25  RelationshipSatisfaction            1470 non-null   int64
26  StandardHours                      1470 non-null   int64
27  StockOptionLevel                   1470 non-null   int64
28  TotalWorkingYears                  1470 non-null   int64
29  TrainingTimesLastYear              1470 non-null   int64
30  WorkLifeBalance                    1470 non-null   int64
31  YearsAtCompany                     1470 non-null   int64
32  YearsInCurrentRole                 1470 non-null   int64
33  YearsSinceLastPromotion             1470 non-null   int64
34  YearsWithCurrManager                1470 non-null   int64
dtypes: int64(26), object(9)
memory usage: 402.1+ KB
```

In [8]:

df.describe()

Out[8]:

	Age	DailyRate	DistanceFromHome	Education	En
count	1470.000000	1470.000000	1470.000000	1470.000000	14
mean	36.923810	802.485714	9.192517	2.912925	1.0
std	9.135373	403.509100	8.106864	1.024165	0.0
min	18.000000	102.000000	1.000000	1.000000	1.0
25%	30.000000	465.000000	2.000000	2.000000	1.0
50%	36.000000	802.000000	7.000000	3.000000	1.0
75%	43.000000	1157.000000	14.000000	4.000000	1.0
max	60.000000	1499.000000	29.000000	5.000000	1.0

8 rows × 26 columns



In [9]:

df.columns

Out[9]:

Index(['Age', 'Attrition', 'BusinessTravel', 'DailyRate',
'Department',
'DistanceFromHome', 'Education', 'EducationField',
'EmployeeCount',
'EmployeeNumber', 'EnvironmentSatisfaction', 'Gender',
'HourlyRate',
'JobInvolvement', 'JobLevel', 'JobRole', 'JobSatisfaction',
'MaritalStatus', 'MonthlyIncome', 'MonthlyRate',
'NumCompaniesWorked',
'Over18', 'OverTime', 'PercentSalaryHike',
'PerformanceRating',
'RelationshipSatisfaction', 'StandardHours',
'StockOptionLevel',
'TotalWorkingYears', 'TrainingTimesLastYear',
'WorkLifeBalance',
'YearsAtCompany', 'YearsInCurrentRole',
'YearsSinceLastPromotion',
'YearsWithCurrManager'],
 dtype='object')

```
In [10]: df.isnull().sum()
```

```
Out[10]: Age                                0
Attrition                                0
BusinessTravel                            0
DailyRate                                0
Department                                0
DistanceFromHome                          0
Education                                0
EducationField                             0
EmployeeCount                             0
EmployeeNumber                             0
EnvironmentSatisfaction                    0
Gender                                    0
HourlyRate                                0
JobInvolvement                             0
JobLevel                                  0
JobRole                                   0
JobSatisfaction                             0
MaritalStatus                             0
MonthlyIncome                             0
MonthlyRate                                0
NumCompaniesWorked                        0
Over18                                    0
OverTime                                  0
PercentSalaryHike                         0
PerformanceRating                         0
RelationshipSatisfaction                    0
StandardHours                             0
StockOptionLevel                          0
TotalWorkingYears                         0
TrainingTimesLastYear                     0
WorkLifeBalance                           0
YearsAtCompany                             0
YearsInCurrentRole                         0
YearsSinceLastPromotion                    0
YearsWithCurrManager                       0
dtype: int64
```

```
In [12]: df['Attrition'].value_counts()
```

```
Out[12]: Attrition
No      1233
Yes      237
Name: count, dtype: int64
```

```
In [13]: df.duplicated().sum()
```

```
Out[13]: np.int64(0)
```

In [14]:

```
# Overall attrition rate

attrition_rate = df['Attrition'].value_counts(normalize=True) * 100
print(attrition_rate)
```

Attrition
No 83.877551
Yes 16.122449
Name: proportion, dtype: float64

In [15]:

```
# Break it down by department
dept_attrition= df.groupby('Department')
['Attrition'].value_counts(normalize=True)
print(dept_attrition)
```

Department Attrition
Human Resources No 0.809524
 Yes 0.190476
Research & Development No 0.861602
 Yes 0.138398
Sales No 0.793722
 Yes 0.206278
Name: proportion, dtype: float64

In [22]:

```
# Does overtime affect attrition?
df.groupby('OverTime')
['Attrition'].value_counts(normalize=True).unstack()*100
```

Out[22]:

Attrition	No	Yes
OverTime		
No	89.563567	10.436433
Yes	69.471154	30.528846

```
In [23]: # Does job satisfaction affect attrition?
df.groupby('JobSatisfaction')
['Attrition'].value_counts(normalize=True).unstack()*100
```

Out[23]:

	Attrition	No	Yes
JobSatisfaction			
1		77.162630	22.837370
2		83.571429	16.428571
3		83.484163	16.515837
4		88.671024	11.328976

```
In [24]: # Does income affect attrition
df.groupby('Attrition')['MonthlyIncome'].mean()
```

Out[24]: Attrition
No 6832.739659
Yes 4787.092827
Name: MonthlyIncome, dtype: float64

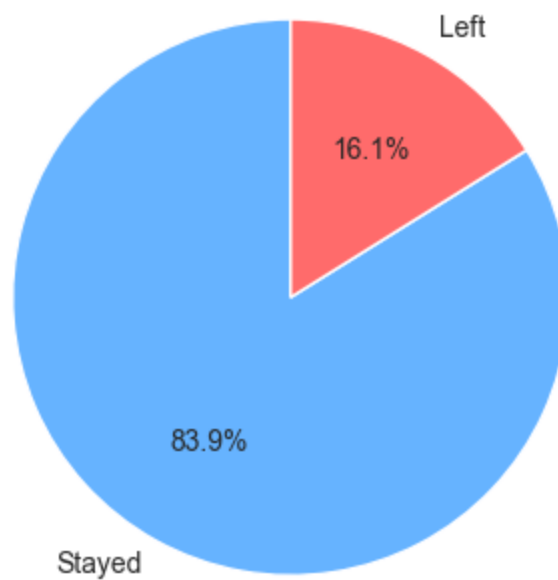
```
In [25]: # Does the year since last promotion affect attrition?
df.groupby('Attrition')['YearsSinceLastPromotion'].mean()
```

Out[25]: Attrition
No 2.234388
Yes 1.945148
Name: YearsSinceLastPromotion, dtype: float64

```
In [27]: sns.set_style("whitegrid")
```

```
In [34]: # Overall Attrition(pie Chart)
plt.figure(figsize=(4,4))
df['Attrition'].value_counts().plot.pie(
    autopct='%1.1f%%',
    colors=['#66b3ff', '#ff6b6b'],
    startangle=90,
    labels=['Stayed', 'Left']
)
plt.title('Overall Employee Attrition Rate', fontsize=14,
fontweight='bold')
plt.ylabel('')
plt.tight_layout()
plt.savefig('chart1_overall_attrition.png', dpi=150)
plt.show()
```

Overall Employee Attrition Rate



```
In [44]: # Attrition by Department (Bar Chart)

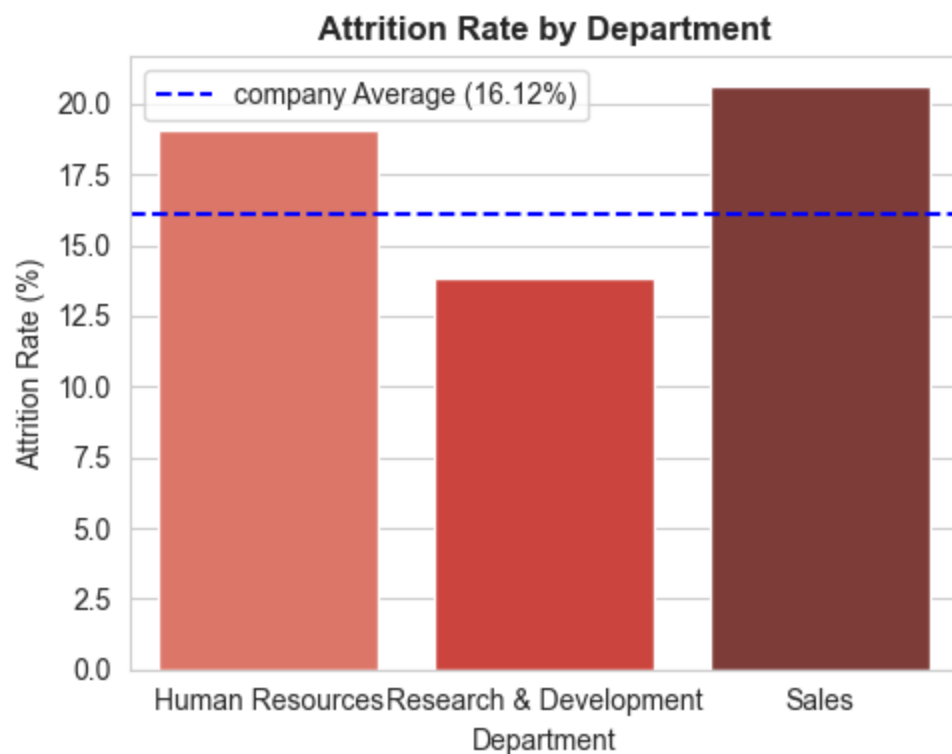
dept_data=df.groupby('Department')['Attrition'].apply(
    lambda x: (x=='Yes').sum()/len(x)*100
).reset_index()
dept_data.columns=['Department','Attrition_Rate']

plt.figure(figsize=(5,4))
sns.barplot(data=dept_data, x='Department',y='Attrition_Rate',
palette='Reds_d')
plt.axhline(y=16.12, color='blue', linestyle='--', label='company Average
(16.12%)')
plt.title('Attrition Rate by Department', fontweight='bold')
plt.ylabel('Attrition Rate (%)')
plt.legend()
plt.tight_layout()
plt.savefig('chart2_overall_attrition.png', dpi=150)
plt.show()
```

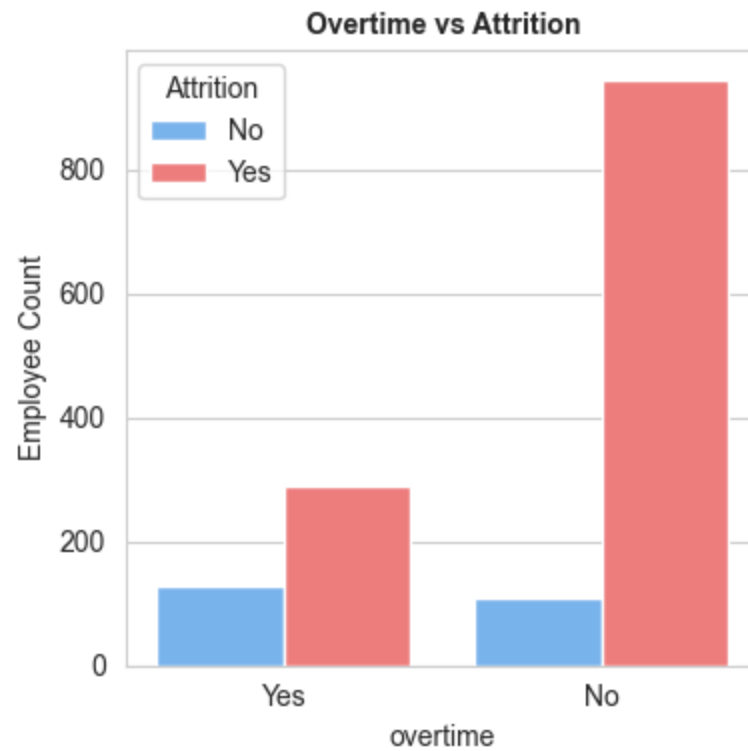
C:\Users\abhis\AppData\Local\Temp\ipykernel_23284\1814889261.py:9:
FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will
be removed in v0.14.0. Assign the `x` variable to `hue` and set
`legend=False` for the same effect.

```
sns.barplot(data=dept_data, x='Department',y='Attrition_Rate',
palette='Reds_d')
```




```
In [51]: # OverTime vs Attrition (Count plot)
plt.figure(figsize=(4,4))
sns.countplot(data=df, x='OverTime', hue='Attrition', palette=
['#66b3ff', '#ff6b6b'])
plt.title('Overtime vs Attrition', fontsize=10, fontweight='bold')
plt.xlabel('overtime')
plt.ylabel('Employee Count')
plt.legend(title='Attrition', labels=['No', 'Yes'])
plt.tight_layout()
plt.savefig('chart3_overtime_attrition.png', dpi=150)
plt.show()
```

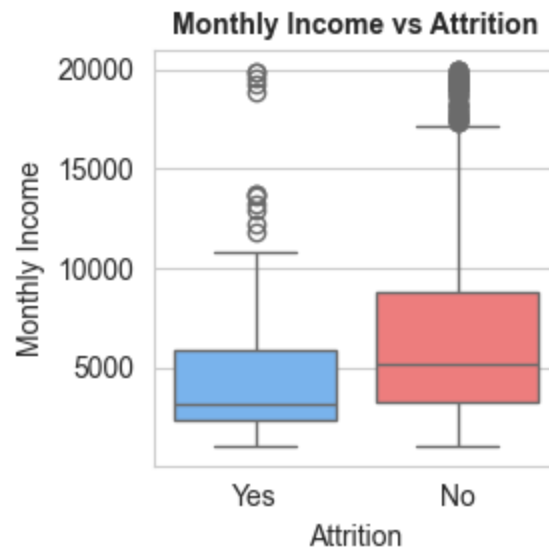


```
In [69]: # Monthly Income VS Attrition(BoxPlot)
plt.figure(figsize=(3,3))
sns.boxplot(data=df, x='Attrition',y='MonthlyIncome',palette=
['#66b3ff','#ff6b6b'])
plt.title('Monthly Income vs Attrition', fontsize=10, fontweight='bold')
plt.xlabel('Attrition')
plt.ylabel('Monthly Income')
plt.tight_layout()
plt.savefig('chart4_income_attrition.png', dpi=150)
plt.show()
```

C:\Users\abhis\AppData\Local\Temp\ipykernel_23284\3245078461.py:3:
FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will
be removed in v0.14.0. Assign the `x` variable to `hue` and set
`legend=False` for the same effect.

```
sns.boxplot(data=df, x='Attrition',y='MonthlyIncome',palette=
['#66b3ff','#ff6b6b'])
```



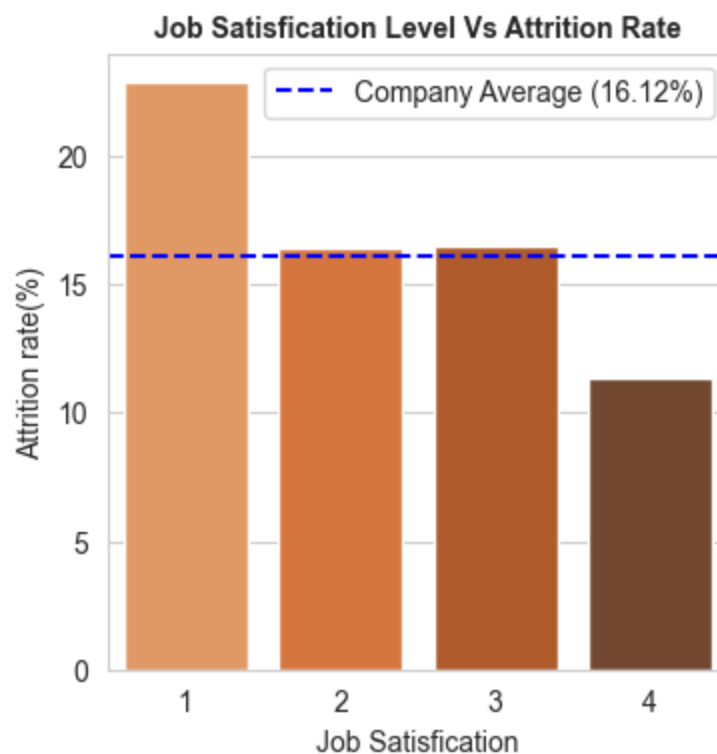
```
In [77]: # Job Satisfaction vs Attrition(BarChart)
js_data=df.groupby('JobSatisfaction')['Attrition'].apply(
    lambda x: (x=='Yes').sum()/len(x)*100
).reset_index()
js_data.columns=['JobSatisfaction', 'Attrition_Rate']

plt.figure(figsize=(4,4))
sns.barplot(data=js_data, x='JobSatisfaction', y='Attrition_Rate',
    palette='Oranges_d')
plt.axhline(y=16.12, color='blue', linestyle='--', label='Company Average
(16.12%)')
plt.title('Job Satisfaction Level Vs Attrition Rate', fontsize=10,
    fontweight='bold')
plt.xlabel('Job Satisfaction')
plt.ylabel('Attrition rate(%)')
plt.legend()
plt.savefig('chart5_jobsatisfaction_attriton.png', dpi=150)
plt.show()
```

C:\Users\abhis\AppData\Local\Temp\ipykernel_23284\3126848017.py:8:
FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will
be removed in v0.14.0. Assign the `x` variable to `hue` and set
`legend=False` for the same effect.

```
sns.barplot(data=js_data, x='JobSatisfaction',
    y='Attrition_Rate', palette='Oranges_d')
```



Key Findings & Business Recommendations

Finding 1: Overtime is the Biggest Attrition Driver

- Employees working overtime have 30.5% attrition vs 10.4% for non-overtime
- That's nearly 3x higher risk
- **Recommendation:** HR should audit departments with high overtime load, introduce compensatory offs or flexible hours, and set a maximum overtime threshold policy. Flag employees with 3+ consecutive months of overtime for manager check-ins.

Finding 2: Low-Income Employees Leave More

- Employees who left earned an average of ₹4,787/month vs ₹6,832 for those who stayed — a 30% income gap
- **Recommendation:** Conduct a salary benchmarking exercise against industry standards, especially for entry and mid-level roles. Introduce structured annual increment policies tied to performance.

Finding 3: Sales Department Needs Attention

- Sales has the highest attrition at 20.6%, above the company average of 16.12%
- **Recommendation:** Review sales incentive structures, target pressures, and onboarding quality. Consider stay interviews with Sales employees in their first year.

Finding 4: Job Satisfaction Matters — But Not Linearly

- Satisfaction Level 1 (lowest) shows 22.8% attrition
- Satisfaction Level 4 (highest) shows only 11.3% attrition — half the rate
- **Recommendation:** Introduce quarterly pulse surveys to track satisfaction early. Build manager training around recognition and feedback culture.

Finding 5: Promotion Delay is NOT a Strong Driver

- Years since last promotion showed almost no difference between those who left (1.94) vs stayed (2.23)
- This is an honest finding — not every assumption holds in data, and acknowledging weak signals shows analytical maturity.

In []:

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